ActInf Textbook Group ~ Cohort 3 ~ Meeting 6 (Chapter 3, part 1)

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hello and welcome everyone
it's March 1st 2023
we're in cohort three
and we are having our first discussion
today on chapter three
chapter three
um is the high road to active inference
previously in chapter two we talked
about the low road and today we're
talking about the high road
those two roads are laid out in figure
1.2 we took it from Bayes theorem to
active inference in chapter two and
chapter three is going to be starting
from the free energy principle and also
taking us to octave inference so before
we go through the chapter or look at any
questions does anyone want to just give
any reflection or thought
what was
some experience they had in reading
chapter 3
or a quote or an idea
or something that stood out for them
you can raise your hand or just go for
it
okay so feel free to raise your hand or
write in the chat if you ever want to
add anything
um
chapter three takes a different attack
than chapter two
[Music]
anyone want anything or we'll we'll
start

exploring through the chapter and then
coming to the questions that people have
raised
um today or in the next week
okay
we'll move through chapter three
hopefully
relatively quickly
to get to the first of the questions and
then we'll have seeded some questions
for coming to our discussion next week
so in section 3.1
in chapter 2
free energy was motivated as a bound
in a way that we can do approximate
Bayesian inference
coming from the low road to active
inference and you can do free energy
based or energy-based methods like a
variational autoencoder you could do
that on any Bayesian statistical problem
at all and it's used commonly
it doesn't necessarily mean that it's
connected to action selection at all
in chapter 3 they're going to start from
The High Road
which is this Central imperative that
organisms or really just things
must maintain their existence
which is going to be operationalized in
a way that's compatible with the
surprise minimization that was brought
up in chapter two
and just like in Chapter 2 where free
energy was proposed as a way to bound
surprise which is like what we would
really want to know
coming from the high road we also want

to bound surprise and it turns out that
minimization of free energy is again
going to appear as a computational
computationally tractable solution to
the problem
the chapter is going to describe the
formal equivalence between the
minimization of variational free energy
and maximization of model evidence or
self-evidencing in Bayesian inference
and then active inference is going to be
brought up because we're going to head
into chapter 4 which is going to be all
about the generative models at the heart
of active inference
active inference is about how organisms
or adaptive systems maintain their
existence by minimizing surprise
via more proximally this tractable proxy
variational free energy
and the two ways to minimize variational
free energy are perception and action
change your mind change the world
integrated imperative
that supports this unifying perspective
on systems
high road is going to start from the
premise
that any living organism or any
persistently observed thing
has to maintain itself in a set of
preferred States
if it doesn't maintain itself in a set
of expected slash preferred States it
just isn't that kind of thing
that is going to apply to everything
from an oil droplet
that's diffusing or not

to
more sophisticated cognitive goals
humans physiological entities have to
stay within their physiological ranges
in reward learning it might be proposed
that there's this value function where
acceptable ranges are more rewarding
and then rewarding Paths of action would
be selected in contrast in active
inference
we establish the expectations around
those physiological variables in terms
of a preference and then we reduce our
surprise about realizing those
preferences slash expectations
that allows us to take the most likely
course of action
to maintain being the kind of thing we
are rather than proposing this reward
function and then asking what's the most
rewarding outcome
foreign
active inference cast the biological
problem of or explanation for survival
as surprise minimization
surely that will spur many conversations
but for now this is not like surprise
birthday minimization so this isn't to
say that surprise doesn't happen or that
cognitive surprise is not even sought
after actively in certain cases this is
referring to the statistical concept of
surprise
which is defined for a very specific
generative model
so not just surprise in general but for
a given generative model
at whatever level it's constructed at

surprise minimization
is what leads to the maximization of
model evidence because if you can
minimize surprise or the tractable proxy
of variational free energy
your model will basically be as fit as
it can be
against any
thoughts or just raise your hand and we
can take it anyway
3.2
mark off blankets
does anyone want to just give a
preemptive thought on Marco blankets or
anything else Jonathan go for it
sorry it's actually with with respect to
the to the surprise and and this is sort
of with respect to some preference
distribution over the the status of the
um of the agent
presumably does the agent have to have
any awareness of that preference
distribution itself and can the
preference distribution itself be
something which is inferred
um you know we may infer something about
our our preferred States
um you know uh is there a sort of
hierarchy to that as well
nice questions does anyone want to give
a first thought does the agent need
awareness of preference and can
preferences be learned
okay just first passes uh Ali go for
uh yeah actually I believe uh it can be
modeled
um in a kind of uh nested hierarchical
way so that uh the agent can be aware of

the aware of its own preferences and
also preferences uh is not uh static at
all and they can be dynamic as well and
can be learned and updated
great exactly if the model
does have a higher nested level
that
enables the interpretation of awareness
on preferences sure
in the base model no no awareness
attention phenomenology qualia is ever
implied at the bazel level so it doesn't
have to be but you could make a model
that has that feature can preferences be
learned again in the most simple case no
the C Vector is fixed however there have
been several works by Sajid at all and
others on preference learning
will
oh I feel like you said most of what I
was just about to say just that you know
thinking at the level of a bacterium you
know awareness is a a tricky concept but
um clearly there is an embedding of
these preferences in in you know some
information at whatever level we're
talking about
and yeah talking about learning
preferences you know perhaps at the
highest level those don't really change
but if you're talking about lower level
preferences
um those are kind of adapted to to the
individual uh time span like the
bacteria may not
um kind of change its Behavior overall
on on the on a lifespan but uh you know
given an individual couple of minutes

it's going to say oh I prefer to go left
versus right because of the chemical
gradient
great yes so
a preference slash expectation for being
in the Run versus the tumble mode of
movements
is happening faster and might be
learnable or updatable whereas a slower
updating process of for example whether
sugar is preferred at all might be
happening at a different time scale
um ever and Francois
yes I'm also thinking about Michael
Levin's work you had him on one of your
um episode former episodes where he's
talking about basal cognition
and that there's maybe even something
like nowhere of preferences at the very
very lowest level
cool
thank you Francois and then Jonathan
and the example that that will said that
you know lower level preferences could
could change like moving left or right
for the bacterium
how do we know if that's a preference or
not just like it's re-evaluating
different policies that would lead to
different expected free Energies
yeah it's a great question
if we don't Define the system like it's
not a system that we created then it
would be an empirical question whether
the data support a model in which you
have a fixed preference
with oscillating policy selection based
upon expected free energy calculations

or whether there are behavior that seems
to go beyond that to actually capture
uh a plastic C
so it's an empirical question from a
given data set and from a given
because we can easily imagine and just
speak out both of those examples and so
it's like a portfolio of models and then
we would do model selection or
evaluation amongst those alternative
strategies
um Jonathan and then Evert
yeah I mean is there some
um some way of thinking about it that
there's always going to be a hierarchy
that there is simply a preference for
survival
within within any model like that that
um you know the agent is trying to
retain its thickness
um and from that maybe glucose levels
and temperature and you know what
whatever else it may be but each of
those is a preference in sort of uh
in order to have this macro preference
which is simply survival
that's a great way to put it over
intra and especially intergenerational
time scales
you're only going to ever see things
that at least act that way or appear
that way
that's kind of like the good regulator
theorem from cybernetics
so yeah at some point
it can be said that there's this
implicit
pervasive preference

for persistence
otherwise you just wouldn't be even
observing that system to be anyway at
all
ever
yeah would you say there's a difference
between preferences and
goal directed Behavior
it's a great question in fact goal is
not used in the active inference
ontology we can definitely talk
informally about like
well the preference is to be on Square
three comma 3 in the grid world so just
Loosely speaking its goal is to make it
to three comma three
but we don't need to appeal to goals as
a formal construct nor reward
so
conversationally it's totally fair to
talk about uh agentic or goal directed
Behavior
the way that that's being
operationalized in active inference is
through this preference vector
thanks Scott um always might be off
there are presumably bad Regulators
which maintain themselves by luck and
bad Regulators which will lose their
homeostasis imminently yes
totally agreed with that luck is only
going to go
so far and so long
and the fact that the imperative is for
adaptive regulation doesn't mean there
aren't systems that fail
this is probably also related to what we
discussed before and as if talk that the

agent looks as if it's minimizing this
or that or it looks as if it's pursuing
a certain goal
would you say that's right
yes as if or perhaps a synonym for as if
could be modeled as or I could publish a
paper modeling it as
as a very um
uh defined way of talking about as if I
can imagine it
or it could be written that or one could
model that
and that puts us more squarely in the
instrumentalist perspective or just the
imaginative perspective
so awesome comments
great
from
surprised with respect oh um I'll ego
for it
uh yeah I just wanted to point out uh
perhaps we can talk about gold-directed
motion as a kind of uh emergent uh
phenomena that arises from this
energy minimization mechanism so it's
not by itself an objective parameter
that we need to describe or model it's
just uh it can be
seen or described as an emergent
phenomenon of that mechanism
awesome
and I just uh Everett go for it sorry I
just had to like re-share my screen
I'm sorry I had sorry for my handset all
right
so continuing on with Section 3.2
big topic
in the field

the way that
things are going to be modeled or framed
is in terms of their relative insulation
from the environment
the generative process the environment
and the relative insulation with the
generative model or the agent the
particular States
in the absence of the separation there
would be no surprise to minimize there
must be something to be surprised and
something to be surprised about
Unity is plural at minimum two
in other words there are at least two
things system and environment and these
can be disambiguated from one another
these can be modeled as being
disambiguated from one another
a formal way to express that separation
is the statistical construct of a Markov
blanket
which can be traced to the work of Judea
Pearl on Bayesian graphs and causal
here
is box 3.1 with a technical definition
of a Markov blanket
a Markov blanket in brief they write is
a set of variables that mediate all
statistical interactions between a
system and its environment so Markov
blankets are nodes one or more nodes on
a Bayesian graph
for which two other sets of nodes are
conditionally independent
so if person a says something to person
B and B says something to C
it's like person B is the blanket
between a and C

so it isn't the internal external States
can't influence each other but rather
it's that their engagements with each
other are mediated through the blankets
and um just earlier
uh it's in chapter nine we had a really
great discussion about the map territory
fallacy and the map territory fallacy
fallacy
and for now just to sort of see this
discussion and continue
um the movement
Markov blankets exist for generative
models because Markov blankets are a
feature of Bayesian graphs so the world
the territory does not have Markov
blankets
however
Maps can have Markov blankets
and a huge amount of
well intended
and off-key discussion
has lost touch with that map territory
distinction Markov blankets are about
Maps
because we're talking about partition
States on a base graph
the claim is not that the real world
consists of labeled
things that have internal external sense
and action on them
and then just one more point and again
feel free to raise your hand or anything
in the discussions of pearl
the Markov blanket is this insulating
set of nodes
what friston and collaborators have done
in the previous two decades or so

is
further partition
that undirected Markov blanket into two
types of notes
nodes with incoming dependencies
which are interpreted as sensory States
and nodes with outgoing dependencies
which are active States
that licensed the use of the Markov
blankets to describe the statistical
boundaries of an Adaptive or cybernetic
agents
so the Markov blanket concept itself is
undirected
it just describes the insulating set of
nodes
friston at all operationalized it
further to bring it into the setting of
cognitive modeling
so now we revisit
that particular partition
that previously was shown as a feedback
loop with perception and cognition and
action
and now it's going to be shown in a way
that's going to be taking us closer to
the Bayesian mechanics or the particular
partition which is we have the blanket
which is shown here according to the the
friston blanket approach active states
with outgoing statistical dependencies
sensory states with incoming statistical
dependencies
internal states don't have a direct Edge
to external States so there's no
telekinesis
and there's no telepathy
however internal states can select

actions
that modify how the external World
changes
that influences which sensory states are
observed which is interpreted through
perceptions learning by internal States
so consequences percolate throughout the
entire partition
but there's no direct connections
between external and internal States Ali
uh yeah I just wanted to quickly point
out
um about two issues here one of which is
um I think an important error in this
figure uh because uh well we know that
the flow of active States uh wouldn't
depend on the external States and
similarly the flow of the internal
sensory States wouldn't will not depend
on the internal state so in the
equations in that figure X and Y should
be interchanged yeah that's it and the
other one is
of course to emphasize my favorite
definition of markup blanket which is
the uh it's a kind of boundary or
interface in state space so it's not
necessarily a spatial temporal boundary
although uh in some cases it might
manifest itself as a spatial temporal
boundary such as the membranes of cells
or organelles or whatever
great way to say it the markout blanket
is the boundary in the statistical State
space okay so
um I'm just gonna add some of these
questions uh Jonathan I'm just adding it
to the chapter three questions and we'll

we'll
um come to it
I'm just having this weird thing with
like
freezing of my screen
um and
than ever
yeah I just posted the link in the chat
I don't know if it's possible to open
the page yes
but because there isn't a picture a
little bit down
I think that's also a mark of blanket
that's shown there but there are more
um more arrows than in the one that you
just showed in the textbook
a little bit lower
yeah so if you and I wonder yeah so I
wonder no it's not that one okay which
page
oh I have a look
uh it's on page
let's figure one
okay I'm gonna have to reshare again
sorry
okay I hope that this is a little bit
more stable
the the
uh well just
a few possible comments any base graph
is going to have a multitude of Markov
blankets Markov blankets are not
uniquely identifying like agent
environment boundaries but those are
some boundaries in state space that
we're interested in but every base graph
that isn't fully connected so if there's
any nodes that aren't connected to other

nodes they will have a Marcon blanket so
the concept is actually less
philosophically and even semantically
loaded than people often expect any base
graph you just draw on a piece of paper
if there's some node that intermediates
two other sets of nodes
it's a markup blanket it does not
identify cybernetic system boundaries
with respect to the cybernetic system
boundary though which is one kind of
base graph that we care about
the work of Aguilera at all how
particular is the free energy principle
has explored different topologies of the
particular partition
for example like what if you do or don't
have this Edge between active and
sensory States why is there an arrow
going from action back to internal why
is there an arrow going from sense back
to external
the first answer on this
is that this is a statistical model so
just because you're enabling that edge
doesn't mean that the data are going to
support that edge having a non-zero
parameterization
so just describing the possibilities
that are going to get parameterized so
it's not a claim about the real world
#map territory it's just saying this is
the class of models that were fitting
so in practice
it means less
than it may seem however it's also the
case that what constraints we put on
this topology do end up influencing

greatly what kinds of models we fit
Michael
yeah I have a question on page 43 under
this box it says
illustrates the interpretation of marker
blanket in a dynamic setting and then
the sentence here the conditional
independencies have been supplemented
with dynamical constraints so that the
flow do not depend on states on the
opposite opposite side of the blanket so
can you elaborate what that means and
what this implies uh What uh dynamical
constraints in contrast to
what else
the the dynamical constraints are the
dynamical equations that are provided
for the rates of change of each of these
and again note what um what Ali has
brought up in and written in the chat
um there there's just a slight error
with the the variables as written
so where it says that U Dot
so that's like action States
it says that it's a function of X U and
Y
um it should be mu U and Y
because this would suggest that octave
states do depend on external States when
actually this x should be a mu
which is to say that active States
are a function of internal States
sensory states and their own current
position
that doesn't mean that they're all equal
contributors it's empirical with your
data set and your generative model
and you can imagine situations where

it's only the internal state that
matters or it's only the sensory state
that matters this is just like the space
in which models are fit from
this isn't to Claim about
any specific cybernetic system or any
interpretation of the real world
octopus
uh how so x-star indicates that you're
dealing with continuous time Dynamics
correct these are flows on the variables
Okay so
um I'm having a tough time mapping this
is there any disjunction or any
difficulty with things that are
inherently discrete
uh like I love cellular automata and
any discrete agent-based model can be
can be put in a cellular automaton
framework but there's no way to really
I mean you can I guess you can
interpolate it somehow but it's not a
natural way of thinking about it so is
continuous time essential to this
framework or is it just a convenient
much of the textbook thank you octopus
is about the differences and
similarities between discrete time and
continuous time formalizations so just
to give a quick preview figure 4.3 shows
on the top The partially observable
Markov decision process discrete time
discrete State space formalization and
the bottom is the continuous time
formalization which is more like a
Taylor series approximation using the
generalized coordinates of motion
and then chapter 7 focuses on the

discrete time generative models chapter
8 focuses on the continuous time models
and in figure 8.6 there's a nested
hierarchical model where the lower level
is continuous time
and the higher level is discrete time
generative models can be either discrete
or continuous State spaces time or other
variables
uh Jonathan wrote presumably in reality
all these variables are high dimensional
vectors for sure it could be as simple
as a scalar a single number or it could
be an n-dimensional tensor
ever yeah thanks octopus efforts are
Henry's or or anyone else
no I didn't have my hands okay it was
just my zoom was locking okay
is going to give an intuition for what
it means when we have this um
a simple system with mu
internal x external and B blanket States
so we can imagine that we're sampling
from this Cube X Y and Z but it's mu B
and uh and X
and then these three panels
are X conditioned on B mu conditioned on
B and then finally the relationship
between X and mu
and so this shows that you could
estimate
X
conditioned upon an expectation of mu
given B so internal States
given blanket States that's like
cognition
about X so here's my estimate my
posterior probability estimate of X

which I'm not directly observing
conditioned upon
my expectations
of internal States conditioned on B
and so you can have a relationship
between X and B
like there's some correlation between
external and blanket States some
correlation between the temperature in
the room and your thermometer
and then there's some correlation
between your cognitive States and the
if your beliefs about temperature given
the thermometer
and then that results
in there being a an information
relationship between external and
internal States though they are only
mediated through the blanket
measurements
this is just an inference passive
setting there's no action coming into
play yet it's just to get a bit of an
intuition about how a Markov blanket
mediated perceptual inference occurs
3.3 surprise minimization and
self-evidency
so an agent with Markov blanket appears
to model the external environment
in the sense that internal States
correspond on average to a probabilistic
representation
of external states of the system
figure 3.2 that's what that was showing
importantly the agent's generative model
cannot simply mimic external Dynamics
otherwise the agent would simply follow
external dissipative Dynamics so passive

might lead one to want a model where we
simply
estimate external States as well as we
can we just learn optimally
and just sit and watch the show
rather the model must also specify the
preferred conditions for the agent's
existence
or the regions of states that the agent
has to visit to maintain its existence
or satisfy criteria for its existence in
terms of occupying characteristic States
so if we want to do more than just watch
if we want to act
to bound our surprise and improve our
persistence in a dissipative far from
equilibrium environments
we need to have some preferences or
expectations that actions can be
selected
based upon how they will realize those
preferences Francois
yeah sorry I think we kind of skipped
over something but I didn't really
understand we want to ask about
um is that
the thing about the the Symmetry across
the market plan Mark of blanket and the
consequences of this being weakened
um associate pairs of expected internal
and external states with each other I
really don't understand how that
consequence comes from the Symmetry
question mark of blanket could
and could you explain it better
in a different way
sure there's more
um angles let's just say first starting

with a homeostatic angle let's just say
that temperature really is relevant for
so the color of the room doesn't matter
for survival but the temperature matters
and we have a thermometer so that's our
Markov blanket and we're going to be
getting some
proxied information about temperature
through our thermometer
without going too deep into debates
about representationalism
there must be something in the
generative model the agent the cognitive
model of the agents
that represents that temperature there
has to be some mapping
between
the temperature in the room unobserved
and the agent's perception of
temperature such that it's able to
engage adaptively
otherwise it's just going to be a
particle adrift
so it turns out it doesn't need to have
an exact structural relationship
so temperature could be a continuous
variable and the agent's generative
model might just be too hot just right
or cold three states so I don't need to
have the same structure but there does
have to be a mapping
between decisions that influence vital
and the external reality of those vital
variables so that's the homeostatic
angle and then just shortly the other
symmetry based angle is um the external
states don't have to be an active agent
but they could be this could be like a

Communications interface
in which case it would be fair to say
that the generative model performs
inference with an aboutness of the
generative process but also the process
as agent is doing inference on the
aboutness of the agent on the right side
so that has been more recently explored
with like a symmetry with a niche doing
inference on the organism
also through the same blanket
okay thank you that makes more sense
great so that um baking in of
or expectations
builds in what they call an implicit
optimism bias
which is like
steering our perception and action
towards realizing preferences about
observations
this optimism bias is necessary for the
agent to go beyond the mere duplication
of external Dynamics or resemblance of
external Dynamics
to prescribe active states that
underwrite its preferred or
characteristic States
one can cast optimal Behavior with
respect to Prior preferences
which again doesn't mean that it always
works doesn't mean everything always
lives all the time or that outcomes are
the best that they can ever possibly be
but given prior preferences
the maximization of model evidence by
perception is a signal processing
perspective that's like the optimal
reconstruction of some sound

however we have a unified imperative
that isn't just maximizing model
evidence for perception
we have a joint imperative for the
maximization of model evidence by
perception and action
we can reduce our surprise bound our
with minimizing variational free energy
minimizing expected free energy
which is a dual or unified imperative
that brings together changing your mind
through perception learning and changing
the world through action
um a good fit indicates the model
successfully accounts for Sensations
at the same time it realizes its
preferred sensation
given that they are less surprising
good description
good prescription
good scientist
good engineer
such a good fit is a guarantee of
surprise minimization as maximizing
model evidence P of Y the probability of
the data given the model
is mathematically equivalent to
minimizing surprise fancy I
negative natural log of the same
that is why you'll often hear adaptive
systems described as self-evidising
self-evidencing means acting to Garner
sensory data consistent with an internal
model
hence maximizing model evidence
surprise minimization as a hamiltonian
principle of least action
this is starting to

invoke
some more physics based approaches on
the low road we were squarely within
statistics Bayes theorem base theorem
with action active inference
coming from the free energy principle on
the high road
we have a lot more physics based views
expressing this as a physicist might
dot dot
this is like a ball rolling down a hill
from a high gravitational potential
energy at the top of the hill to low
energy in a basin
so just like in classical mechanics it
might describe a ball rolling down
an elevation landscape with respect to
potential energy
we're in an information geometric
landscape and the ball rolling down the
hill taking the path of least action
is going to be formally equivalent or at
least formally analogous to our belief
distributions rolling down the hill
with negative log evidence or surprise
playing the role of gravity
that is what is equivalent to potential
energy drawing the ball down the hill
and if you want a more technical angle
check out live stream 52 which will be
happening this week and next week
it is going to go a lot deeper into this
surprise as potential energy
interpretation
here's a path taken by a two-dimensional
random dynamical system
it's kind of vibrating but it is also
spending most of its time in the sensor

and the left one
another way to view like the landscape
that this pen was drawing on or that
this ball was rolling on is on the right
this is like a topo map
where here is like a plateau and then
it's like a bowl
so even though it started here it kind
of vibrated around but it spends most of
its time in the bottom of the bowl
in the middle
is a trajectory of a system
that has Dynamics bearing no
relationship to surprise
so here you have a uncorrelated just
random walk Brownian diffusion
inner particle
more details and equations
that connect
minimization of entropy
which is kind of like the
um blurriness of a distribution like how
broad or sharp it is
has a relationship with the maximization
of the probability of observations
and also
the expectation of self-information so
ensuring that a small proportion of
sensory states are occupied with high
probability
like we want a distribution on
temperature States
that's very sharp
that is equivalent to bounding surprise
about temperature yes and also
maintaining a certain entropy on the
temperature distribution
Maria does anyone know if the random

pattern in the middle is less or more
common than the centered one Jonathan
wrote it depends on the context totally
agreed
you can add more details than that Maria
but there are um sets of variables in in
your imagination or in the world
for whom their Dynamics look like that
and there's sets that look like that and
various other things in between
surprise minimization formalizes the
idea of homeostasis
uh Ali go for it
uh yeah I think it might be useful to
point out uh Paul Curry's
the taxonomy of singular points because
uh back in the 19th century he showed
that uh singular points can have only
four possible different types uh namely
nodes saddle points Foci and centers but
then he went out uh claiming that uh
among those different types nodes saddle
points and Foci are much more generic
than Center so centers only arises in
exceptional circumstances so
yeah in almost all the physical systems
that we model physical dynamical systems
uh we will most likely encounter only
those three kinds of singular points
um and it's just very briefly uh nodes
are the points through which an infinite
number of solution curves pass saddle
points are the points through which only
two solution curves pass and Foci are
the points which the solution curves
approach in the manner of the
logarithmic spiral and the centers are
the points around which the solution

curves are somehow Inc closed and
enveloping one another
great just really quickly to efforts
uh question so this sensory active state
notice that the notation is totally
different so you got to take a fresh
look at the notation every time like
external states are Q Prime and so on
um so m is a blanket M and Phi are
blanket between q and E
but also e and Omega are a blanket
between q and M so mark all blankets are
everywhere they do no philosophical work
they're a purely technical description
of nodes on a Bayesian graph
maps have Markov blankets not
territories it can be somewhat
misleading when we have a picture of a
brain as if we were describing it but
it's like actually a map of the brain
not the territory itself so brains don't
have Markov blankets Maps do
maps are made by people
maps are made amongst portfolios of
other maps and some of those Markov
boundaries in Phase space
can be identified with different
functional phenomena
but
just drawing a base graph
is not a prediction or an explanation by
itself and so it doesn't do too much
work
yet a huge amount of discourse has
focused on those types of topics ever
yes do you do you agree that that's a
big description of the Marco blanket
that captures more uh maybe things that

have
been discussed in previous uh
times about preferences and subjective
feelings
because I know that Mark songs has
talked about precision as being related
to the things that we consciously
receive as feelings
is that captured here in this picture do
you see
preferences are not shown here unless
one uses preference as a subtype of
internal State
marks Holmes indeed has discussed
extensively Precision affect
Consciousness awareness and so on
in the end it comes down to just the
full specification of the generative
model what data and what phenomena are
being tried to be explained by that
model and data together and just drawing
any kind of um
diagram
is just the beginning of a graphical
abstract or a schema
it's not an explanation prediction
design imperative or control system
it's just a drawing
all right
in our last
five minutes
topics really interesting
so this is our first discussion on
and we got about halfway through we're
just going to peep ahead
and then next week we will
um pick up with the chapter three
questions already there's a bunch of

them
we've been talking about entropy
self-suprisal model evidence
maximization and their relationships
3.4
relations between or among depending on
how you prefer your English
inference cognition and stochastic
Dynamics
table 3.1 is going to juxtapose
statistical physics
Bayesian inference and information
Theory
and a cognitive interpretation
where in rows
there are going to be sort of
three phases
so across columns
these three Fields as the authors are
showing or talking about something
that's basically like the same or
resonant with
box 3.2
is talking about free energy in
statistical physics and active inference
they want to be clear that variational
free energy which is a used commonly in
machine learning
is not necessarily referring to for
example the Gibbs free energy the actual
liberatable energy of the particles
constituting that embodied entity
3.4.1 variational free energy model
evidence and surprise
here surprise
the information on the data given the
is the negative log of model evidence
and model evidence is always less than

the variational free energy which we'll
go into more
3.42 expected free energy and inference
of most likely trajectories so
variational free energy is real time
it's about the model right now and the
data right now
expected free energy is going to be
about likely courses of actions action
sequences are called policies
and that's going to entail the
description of observations that haven't
actually happened yet
Jack
oh yeah I just wanted to ask a question
regarding the distinction between Gibbs
free energy and variational free energy
as you just mentioned in box 3.2 so
um is this just a metaphor between the
two like an analogy or is there a
mathematical like relationship that we
can write down between gives free energy
helmholes for energy and variational
free energy
yes there are formal relationships among
structurally
that that license the description of all
of those as energy
authors like Alex Kiefer
have um
argued Beyond where many go
to talk about the um
uh
actual uh free energy
minimizing features of
Gibbs maintaining systems but let's
return to that
and then 3.5 active inference and novel

Foundation to understand behavior and
plain text section 36 models policies
and trajectories
plain text
reconciliation of inactive cybernetic
and predictive theories under active
active inference emergence of life to
agency plain text
3.9 summary
living organisms visit characteristic
States active inference
thank you everybody
we will return
next week
or in the office hours and just write
your questions in the
relevant pages and we'll continue the
discussion so thanks a lot everybody for
joining
farewell